MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Plot the constraints. Compare the ratios.
Solve with absolute precision.

Geometric Tri-State | Graphical Intersections | Substitution | Elimination | Word Models

A student-first Maths Manthan conceptual and visual revision resource



What This Handbook Includes

- Standard form criteria and geometry of simultaneous lines
- The Tri-State system: intersecting, coincident, and parallel
- Graphical solutions and coordinates of enclosed triangles
- Algebraic solving: Substitution and Elimination workflows
- The high-speed Symmetric Coefficients addition shortcut
- Parameter k consistency protocols and real-world models

Pedagogical Strategy: Foundations First

A linear equation in two variables represents an infinite locus of collinear points. A simultaneous system represents intersecting constraints:

Unique Solution (x_0, y_0) $\iff$ **Intersecting Lines**
Ratio Test: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ $\iff$ **Distinct Slopes**

Always inspect the coefficient ratios before deciding the solving technique!

CBSE 2026–27 Curriculum Verification & Rationalized Scope

Official Scope (Prescribed & Tested): Pair of linear equations in two variables and graphical method of their solution, consistency/inconsistency. Algebraic conditions for number of solutions. Solution by Substitution and by Elimination. Situational word problems.

Excluded / Deleted Topics (Non-Examinable):

- **Cross-Multiplication Method:** Completely deleted from the NCERT syllabus.
- **Equations Reducible to Linear Form:** Systems requiring $u = \frac{1}{x}, v = \frac{1}{y}$ substitutions are deleted.

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 1 | Foundations: Standard Form & Cartesian Geometry

1.1 Definition and General Form of a Linear Equation

An equation which can be put in the form $\mathbf{ax + by + c = 0}$, where $a, b, c \in \mathbb{R}$ and a, b are not both zero ($a^2 + b^2 \neq 0$), is called a **linear equation in two variables** x and y.

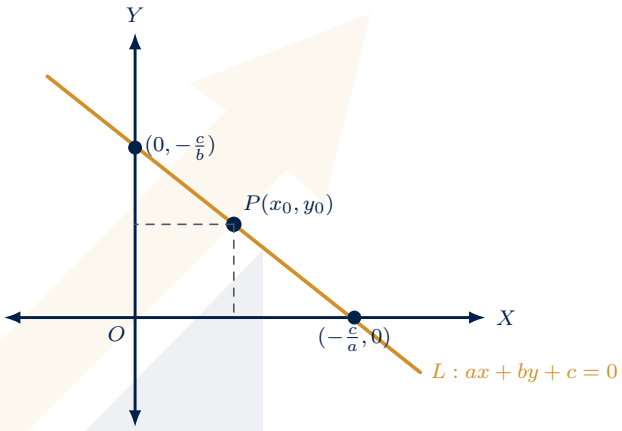
General Form of a Pair of Linear Equations:

$$\mathbf{a_1x + b_1y + c_1 = 0}$$

$$\mathbf{a_2x + b_2y + c_2 = 0}$$

where $a_1^2 + b_1^2 \neq 0$ and $a_2^2 + b_2^2 \neq 0$.

- **Geometric Representation:** Every linear equation in two variables represents a **straight line** on the Cartesian plane.
- **Meaning of Solution:** An ordered pair (x_0, y_0) is a **solution** of the system if and only if it satisfies both equations simultaneously.
- Geometrically, (x_0, y_0) is the **common point** lying on both lines!



1.2 Verifying Whether an Ordered Pair is a Solution

To test whether a given point (x_0, y_0) is a solution of a system, substitute $x = x_0$ and $y = y_0$ into **both equations**. If both LHS equal RHS, the point is a solution; if it fails in even one equation, it is **not** a solution.

Worked Archetype 1 | Algebraic Solution Verification

Problem: Check whether $x = 3, y = 2$ and $x = 1, y = 4$ are solutions of the system:

$$2x + 3y = 12 \qquad \text{and} \qquad x - 2y = -1$$

Check for $(3, 2)$**:**

- Equation 1: LHS $= 2(3) + 3(2) = 6 + 6 = 12 =$ RHS. (Satisfied!)
- Equation 2: LHS $= 3 - 2(2) = 3 - 4 = -1 =$ RHS. (Satisfied!)

Since $(3, 2)$ satisfies both equations, $(\mathbf{3, 2})$ IS a solution of the system.

Check for $(1, 4)$**:**

- Equation 1: LHS $= 2(1) + 3(4) = 2 + 12 = 14 \neq 12$.

Since $(1, 4)$ fails in Equation 1, $(\mathbf{1, 4})$ is NOT a solution.

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CHAPTER 03

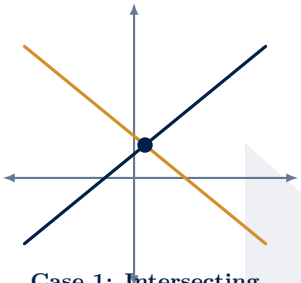
CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

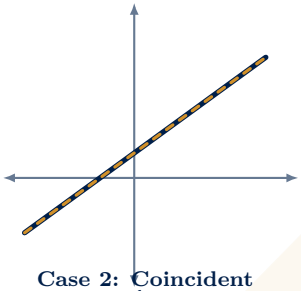
Module 2 | The Tri-State Geometric System: Intersecting, Coincident & Parallel Lines

2.1 The Three Geometrical Configurations of Two Lines in a Plane

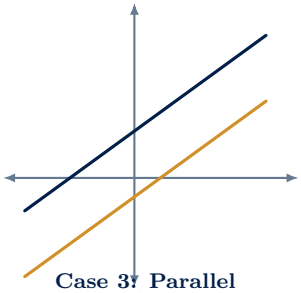
Two straight lines L_1 and L_2 drawn on the Cartesian plane can exhibit exactly one of three spatial relationships:



Case 1: Intersecting
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
Unique Solution
Consistent



Case 2: Coincident
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
Infinitely Many
Consistent (Dependent)



Case 3: Parallel
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
No Solution
Inconsistent

2.2 The Master Ratio Comparison Table

Let $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ be the system in standard form:

Ratio Comparison	Graphical Representation	Algebraic Solutions	System Consistency Status
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting lines	Exactly **one unique solution**	**Consistent** (system has a solution).
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident (overlapping) lines	**Infinitely many solutions**	**Consistent (Dependent).**
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel (non-intersecting) lines	**No solution**	**Inconsistent** (no common point).

Worked Archetype 2 | Testing Consistency via Ratio Inspection

Problem: On comparing the ratios $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$, and $\frac{c_1}{c_2}$, determine whether the following pair of linear equations is consistent or inconsistent:

$$\frac{3}{2}x + \frac{5}{3}y = 7 \qquad \text{and} \qquad 9x - 10y = 14$$

Step 1: Write both equations in standard form $ax + by + c = 0$:

$$\frac{3}{2}x + \frac{5}{3}y - 7 = 0 \implies a_1 = \frac{3}{2}, \quad b_1 = \frac{5}{3}, \quad c_1 = -7$$

$$9x - 10y - 14 = 0 \implies a_2 = 9, \quad b_2 = -10, \quad c_2 = -14$$

Step 2: Compute individual ratios:

$$\frac{a_1}{a_2} = \frac{3/2}{9} = \frac{3}{18} = \mathbf{\frac{1}{6}}, \qquad \frac{b_1}{b_2} = \frac{5/3}{-10} = \frac{5}{-30} = \mathbf{-\frac{1}{6}}$$

Step 3: Compare ratios: Since $\frac{1}{6} \neq -\frac{1}{6} \implies \mathbf{\frac{a_1}{a_2} \neq \frac{b_1}{b_2}}$.
Conclusion: The lines intersect at a unique point. Hence, the system is **consistent**.

CHAPTER 03

CORE SYLLABUS

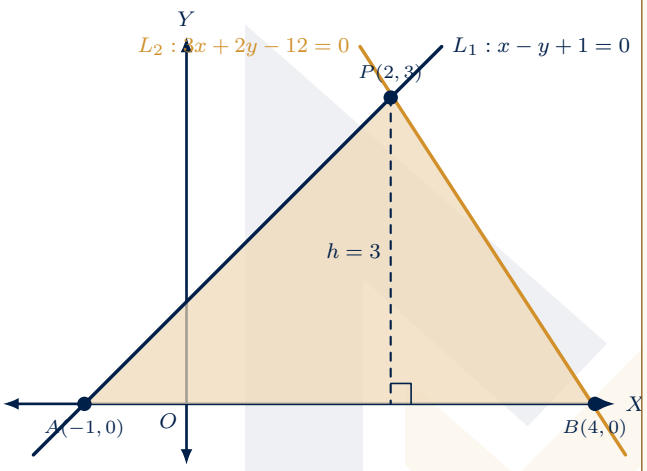
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Module 3 | The Graphical Solution Method & Bounded Triangular Regions

3.1 The 4-Step Graphical Solution Protocol

To solve a system graphically:

1. Express y in terms of x for each equation: $y = \frac{-a_1x - c_1}{b_1}$.
2. Create a table of at least two (preferably three) integer solution pairs (x, y) for each line.
3. Plot the coordinates on a uniform scale and draw straight lines extending across both axes.
4. The coordinates (x, y) of the point where the lines intersect give the unique solution!



Area of the Bounded Triangle Region: When lines intersect each other at $P(x_0, y_0)$ and intersect the x-axis at $A(x_1, 0)$ and $B(x_2, 0)$:

- **Base of Triangle (AB):**

$$\text{Base} = |\mathbf{x_2} - \mathbf{x_1}|$$

- **Height of Triangle (h):** The vertical distance from apex P to the x-axis:

$$\mathbf{h} = |\mathbf{y_0}|$$

- **Enclosed Triangle Area:**

$$\text{Area} = \frac{\mathbf{1}}{\mathbf{2}} \times \text{Base} \times \text{Height} = \frac{\mathbf{1}}{\mathbf{2}} \cdot |\mathbf{x_2} - \mathbf{x_1}| \cdot |\mathbf{y_0}|$$

Worked Archetype 3 | Graphing Lines and Shading the Enclosed Triangle

Problem: Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis, and calculate its area.

Step 1: Solution table for $L_1 : y = x + 1$: Points: $(-1, 0)$, $(0, 1)$, $(2, 3)$.
Step 2: Solution table for $L_2 : y = \frac{12-3x}{2}$: Points: $(0, 6)$, $(2, 3)$, $(4, 0)$.
Step 3: Point of Intersection: Both tables contain the common point $\mathbf{P(2, 3)}$. Hence, $x = 2, y = 3$.
Step 4: Intersections with x-axis ($y = 0$):

- For L_1: $x - 0 + 1 = 0 \implies x = -1 \implies \mathbf{A(-1, 0)}$.
- For L_2: $3x + 2(0) - 12 = 0 \implies 3x = 12 \implies x = 4 \implies \mathbf{B(4, 0)}$.

Vertices of the Triangle: $\mathbf{A(-1, 0)}$, $\mathbf{B(4, 0)}$, $\mathbf{P(2, 3)}$.
Area Calculation: Base $AB = 4 - (-1) = 5$ units. Height $h = 3$ units.

$$\text{Area}(\triangle\mathbf{PAB}) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 5 \times 3 = \mathbf{7.5} \text{ sq. units.}$$

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CHAPTER 03

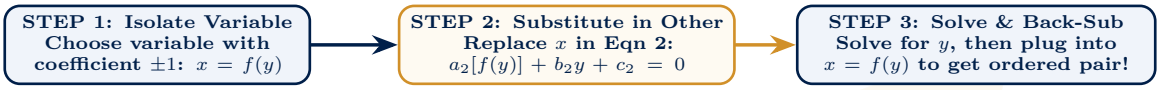
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PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 4 | Algebraic Method I: The Method of Substitution

4.1 Theoretical Mechanism of Substitution

The **Method of Substitution** operates by using one constraint to express one variable strictly in terms of the other, thereby reducing a 2-variable simultaneous system into a **single-variable linear equation**.



4.2 Recognizing Infinite Solutions and Inconsistency Algebraically

When substituting $x = f(y)$ into the second equation:

- **Unique Solution:** Produces a determinate value for y (e.g., $5y = 15 \implies y = 3$).
- **Infinitely Many Solutions:** Variables cancel out leaving a **true mathematical identity** with no variables:

$$\mathbf{0 = 0} \quad \text{or} \quad \mathbf{c = c} \implies \text{System has infinitely many solutions (coincident lines).}$$

- **No Solution (Inconsistent):** Variables cancel out leaving a **false mathematical contradiction**:

$$\mathbf{0 = 8} \quad \text{or} \quad \mathbf{-4 = 10} \implies \text{System has no solution (parallel lines).}$$

Worked Archetype 4 | Stepwise Execution of Substitution Method

Problem: Solve the following pair of linear equations by the substitution method:

$$7x - 15y = 2 \quad \text{— (1)} \qquad \text{and} \qquad x + 2y = 3 \quad \text{— (2)}$$

Step 1: Choose the easiest variable to isolate: In Equation (2), the coefficient of x is 1. Isolate x from Equation (2):

$$x = 3 - 2y \quad \text{— (3)}$$

Step 2: Substitute Equation (3) into Equation (1) (DO NOT substitute back into Eqn 2!):

$$7(3 - 2y) - 15y = 2$$

$$21 - 14y - 15y = 2 \implies 21 - 29y = 2$$

$$-29y = 2 - 21 = -19 \implies \mathbf{y = \frac{19}{29}}$$

Step 3: Back-substitute $y = \frac{19}{29}$ into Equation (3) to find x:

$$x = 3 - 2\left(\frac{19}{29}\right) = 3 - \frac{38}{29} = \frac{3(29) - 38}{29} = \frac{87 - 38}{29} = \mathbf{\frac{49}{29}}$$

Conclusion: The unique solution of the system is $\mathbf{x = \frac{49}{29}}$, $\mathbf{y = \frac{19}{29}}$.

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 5 | Algebraic Method II: The Method of Elimination

5.1 The Elimination Balancing Protocol via LCM Multipliers

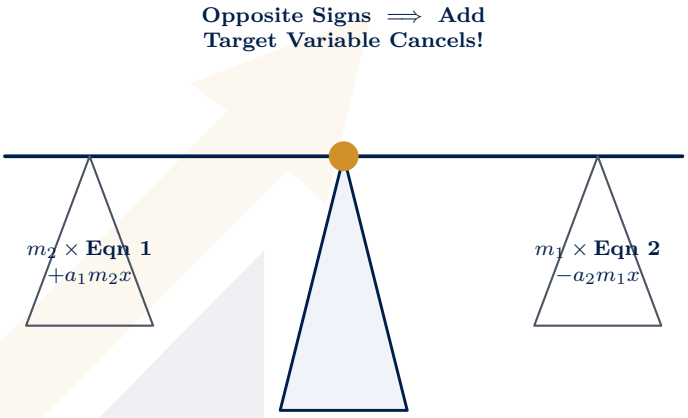
The **Method of Elimination** is generally the fastest and most robust algebraic technique. We multiply one or both equations by non-zero constants so that the coefficients of one variable become numerically equal.

Elimination Operational Rules:

1. Select which variable (x or y) to eliminate based on whichever has the simpler LCM of coefficients.
2. Multiply equations to equalize numerical values:

$$\text{Coefficient of target variable} = \text{LCM}$$

3. **Sign Rule for Elimination:**
   - If coefficients have **Opposite Signs** (+ and −): **ADD** the two equations!
   - If coefficients have **Same Signs** (+ and +, or − and −): **SUBTRACT** the equations!
4. Solve the resulting 1-variable equation and substitute back into any original equation.



Worked Archetype 5 | NCERT Ratio Income / Expenditure Classic

Problem: The ratio of incomes of two persons is 9 : 7 and the ratio of their expenditures is 4 : 3. If each of them manages to save ¤2000 per month, find their monthly incomes.

Step 1: Algebraic Formulation:
Let their monthly incomes be $9x$ and $7x$, and their monthly expenditures be $4y$ and $3y$.

$$\text{Income} - \text{Expenditure} = \text{Savings}$$

$$9x - 4y = 2000 \quad \text{— (1)}$$

$$7x - 3y = 2000 \quad \text{— (2)}$$

Step 2: Eliminate variable y using LCM of coefficients (4 and 3 $\implies$ LCM = 12):
Multiply Equation (1) by 3:

$$3(9x - 4y) = 3(2000) \implies 27x - 12y = 6000 \quad \text{— (3)}$$

Multiply Equation (2) by 4:

$$4(7x - 3y) = 4(2000) \implies 28x - 12y = 8000 \quad \text{— (4)}$$

Step 3: Subtract Equation (3) from Equation (4) (Same signs on $-12y$):

$$(28x - 12y) - (27x - 12y) = 8000 - 6000$$

$$28x - 27x - 12y + 12y = 2000 \implies \mathbf{x = 2000}$$

Step 4: Compute monthly incomes:

$$\text{Income of Person 1} = 9x = 9 \times 2000 = ¤\mathbf{18{,}000}$$

$$\text{Income of Person 2} = 7x = 7 \times 2000 = ¤\mathbf{14{,}000}$$

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

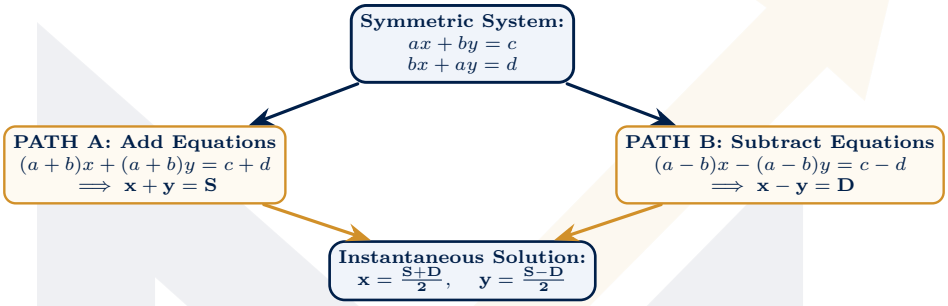
Module 6 | The Symmetric Coefficients Addition/Subtraction Shortcut

6.1 Structural Anatomy of Interchanged Coefficients Systems

When linear systems feature large numerical coefficients where the **coefficients of x and y are interchanged**:

$$\mathbf{ax + by = c} \qquad \text{and} \qquad \mathbf{bx + ay = d}$$

Standard elimination requires calculating huge LCMs (e.g., 152×378), which leads to massive arithmetic blunders. The **Maths Manthan 2-Step Addition/Subtraction Shortcut** reduces the system to mental arithmetic!



Worked Archetype 6 | High-Speed Solution of Large Symmetric Coefficients

Problem: Solve the following pair of equations:

$$152x - 378y = -74 \quad \text{— (1)} \qquad \text{and} \qquad -378x + 152y = -604 \quad \text{— (2)}$$

Step 1: Add Equation (1) and Equation (2):

$$(152 - 378)x + (-378 + 152)y = -74 - 604$$

$$-226x - 226y = -678$$

Divide through by -226:

$$\mathbf{x + y = 3} \quad \text{— (3)}$$

Step 2: Subtract Equation (2) from Equation (1):

$$[152 - (-378)]x + [-378 - 152]y = -74 - (-604)$$

$$530x - 530y = 530$$

Divide through by 530:

$$\mathbf{x - y = 1} \quad \text{— (4)}$$

Step 3: Solve the simplified system of Equations (3) and (4):

$$\text{Add (3) and (4): } 2x = 4 \implies \mathbf{x = 2}$$

$$\text{Subtract (4) from (3): } 2y = 2 \implies \mathbf{y = 1}$$

Conclusion: The solution is $\mathbf{x = 2}$, $\mathbf{y = 1}$. (Solved in 30 seconds without large multiplications!).

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 7 | Consistency Conditions & The Parameter k Protocols

7.1 Rigorous Protocol for Parameter k Determination

A very high-frequency question in CBSE board examinations asks for values of k satisfying a given consistency condition:

Specified Condition	Algebraic Ratio Equation	Solving Protocol	Mandatory Verification Check
Unique Solution	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Solve $a_1b_2 - a_2b_1 \neq 0$	State: "k can be **any real number except** k_0".
Infinitely Many	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Equate Pair 1 ($\frac{a_1}{a_2} = \frac{b_1}{b_2}$) AND Pair 2 ($\frac{b_1}{b_2} = \frac{c_1}{c_2}$)	TRAP ALERT **Both pairs must yield the SAME value of k!**
No Solution	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Equate $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ and verify $\neq \frac{c_1}{c_2}$	Reject any candidate k that makes $\frac{b_1}{b_2} = \frac{c_1}{c_2}$.

Worked Archetype 7 | The Dual-Pair Consistency Trap in Infinite Solutions

Problem: Find the value(s) of k for which the following pair of linear equations has infinitely many solutions:

$$kx + 3y - (k-3) = 0 \qquad \text{and} \qquad 12x + ky - k = 0$$

Step 1: Identify coefficients in standard form:

$$a_1 = k,\ b_1 = 3,\ c_1 = -(k-3) \qquad \text{and} \qquad a_2 = 12,\ b_2 = k,\ c_2 = -k$$

Step 2: Set up the condition for infinitely many solutions:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies \frac{k}{12} = \frac{3}{k} = \frac{-(k-3)}{-k} = \frac{k-3}{k}$$

Step 3: Solve the first pair of ratios:

$$\frac{k}{12} = \frac{3}{k} \implies k^2 = 36 \implies \mathbf{k = \pm 6} \implies k = 6 \quad \text{or} \quad k = -6$$

TRAP ALERT

Critical Board Examination Audit: Do NOT stop here! You must verify which sign satisfies the second ratio:

$$\frac{3}{k} = \frac{k-3}{k}$$

Cross-multiplying (for $k \neq 0$):

$$3k = k(k-3) \implies 3 = k - 3 \implies \mathbf{k = 6}$$

Step 4: Check consistency for $k = -6$:

$$\text{For } k = -6: \quad \frac{a_1}{a_2} = \frac{-6}{12} = -\frac{1}{2}, \quad \frac{b_1}{b_2} = \frac{3}{-6} = -\frac{1}{2}, \quad \frac{c_1}{c_2} = \frac{-6-3}{-6} = \frac{-9}{-6} = \frac{3}{2}$$

Here $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, which gives **no solution**, NOT infinitely many!
Hence, $\mathbf{k = -6}$ is **strictly rejected**.
Conclusion: The only value of k for which the system has infinitely many solutions is $\mathbf{k = 6}$.

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 8 | Word Problems I: Numbers, 2-Digit Reversals & Fractions

8.1 The Two-Digit Number Decimal Place Value Model

A two-digit number cannot be written as xy (which represents a product!). In our base-10 positional numeral system:

$$\text{Original Two-Digit Number} = \mathbf{10x + y} \quad (\text{where } x = \text{tens digit},\ y = \text{units digit})$$

$$\text{Reversed Two-Digit Number} = \mathbf{10y + x}$$

Universal Algebraic Properties:

- **Sum of Number and its Reverse:** $(10x+y)+(10y+x) = 11x+11y = \mathbf{11(x+y)}$ (*always a multiple of 11!*)
- **Difference of Number and its Reverse:** $(10x+y)-(10y+x) = 9x-9y = \mathbf{9(x-y)}$ (*always a multiple of 9!*)

8.2 Fraction Modification Models

Let the original fraction be $\frac{\mathbf{x}}{\mathbf{y}}$ (where x = numerator, y = denominator, $y \neq 0$):

- If c is added to both numerator and denominator: $\frac{x+c}{y+c} = \frac{p}{q} \implies q(x+c) = p(y+c)$.

Worked Archetype 8 | Reversible Two-Digit Number Problem

Problem: The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number. How many such numbers are there?

Step 1: Set up variables: Let the tens digit be x and the units digit be y.
Original Number $= 10x + y$, Reversed Number $= 10y + x$.

Step 2: First condition (Sum is 66):

$$(10x+y)+(10y+x) = 66 \implies 11x+11y = 66 \implies \mathbf{x+y = 6} \quad \text{— (1)}$$

Step 3: Second condition (Digits differ by 2): Since it is not specified whether $x > y$ or $y > x$, we have two cases:

$$\mathbf{x-y = 2} \quad \text{— (2a)} \qquad \text{or} \qquad \mathbf{y-x = 2} \quad \text{— (2b)}$$

Step 4: Solve Case A: Equations (1) and (2a):

$$\text{Adding (1) and (2a): } 2x = 8 \implies x = 4 \implies y = 6-4 = 2$$

$$\text{Number} = 10(4)+2 = \mathbf{42}$$

Step 5: Solve Case B: Equations (1) and (2b):

$$\text{Adding (1) and (2b): } 2y = 8 \implies y = 4 \implies x = 6-4 = 2$$

$$\text{Number} = 10(2)+4 = \mathbf{24}$$

Conclusion: There are **two such numbers**: **42** and **24**.

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

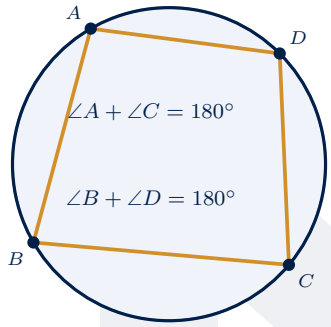
Module 9 | Word Problems II: Age Relations & Geometry / Cyclic Quadrilaterals

10.1 Age Relationship Dynamics: Past vs. Future

Let present ages of two individuals A and B be x and y years:

- n **Years Ago (Past):** Both ages decrease equally: $(\mathbf{x - n})$ and $(\mathbf{y - n})$.
- m **Years Hence (Future):** Both ages increase equally: $(\mathbf{x + m})$ and $(\mathbf{y + m})$.

10.2 Geometric Angle Systems & Cyclic Quadrilaterals



Cyclic Quadrilateral Theorem: A quadrilateral $ABCD$ is cyclic (all 4 vertices lie on a circle) if and only if its **opposite angles are supplementary**:

$$\mathbf{\angle A + \angle C = 180^\circ}$$

$$\mathbf{\angle B + \angle D = 180^\circ}$$

This immediately sets up a decoupled pair of linear equations in x and y!

Worked Archetype 9 | The Angles of a Cyclic Quadrilateral

Problem: $ABCD$ is a cyclic quadrilateral. Find the angles of the cyclic quadrilateral if:

$$\angle A = 4y + 20, \quad \angle B = 3y - 5, \quad \angle C = -4x, \quad \angle D = -7x + 5$$

Step 1: Set up Equation 1 using opposite angles A and C:

$$\angle A + \angle C = 180^\circ \implies (4y + 20) + (-4x) = 180^\circ$$

$$-4x + 4y = 160 \implies \text{Divide by } -4: \quad \mathbf{x - y = -40} \quad \text{— (1)}$$

Step 2: Set up Equation 2 using opposite angles B and D:

$$\angle B + \angle D = 180^\circ \implies (3y - 5) + (-7x + 5) = 180^\circ$$

$$-7x + 3y = 180 \quad \text{— (2)}$$

Step 3: Solve the system using substitution $x = y - 40$ from (1) into (2):

$$-7(y - 40) + 3y = 180 \implies -7y + 280 + 3y = 180$$

$$-4y = 180 - 280 = -100 \implies \mathbf{y = 25}$$

$$x = y - 40 = 25 - 40 \implies \mathbf{x = -15}$$

Step 4: Compute the four angles:

$$\angle A = 4(25) + 20 = 100 + 20 = \mathbf{120^\circ}$$

$$\angle B = 3(25) - 5 = 75 - 5 = \mathbf{70^\circ}$$

$$\angle C = -4(-15) = \mathbf{60^\circ}$$

$$\angle D = -7(-15) + 5 = 105 + 5 = \mathbf{110^\circ}$$

Check: $\angle A + \angle C = 120^\circ + 60^\circ = 180^\circ$, and $\angle B + \angle D = 70^\circ + 110^\circ = 180^\circ$.

CHAPTER 03

CORE SYLLABUS

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 10 | Word Problems III: Fixed Charges & Relative Speed Kinematics

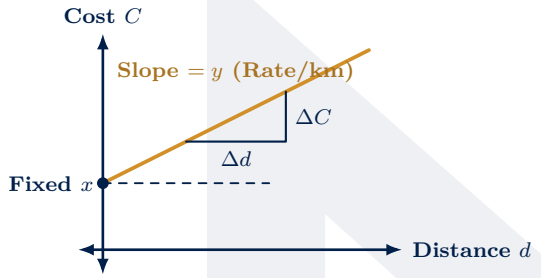
11.1 The Fixed Charge Plus Variable Rate Model

Common real-world models (taxis, library lending, hostel boarding) feature a two-part cost structure:

Total Charge $\mathbf{C(d)}$ = Fixed Base Charge $(\mathbf{x})$ + (Distance / Days $\mathbf{d}$) × Rate per unit $(\mathbf{y})$

$$\mathbf{x + d_1 y = C_1} \qquad \text{and} \qquad \mathbf{x + d_2 y = C_2}$$

Subtracting Equation 1 from Equation 2 immediately eliminates the fixed charge x!



Relative Highway Speeds (Cars A and B): Let car from A have speed x km/h, car from B have speed y km/h $(x > y)$, distance D:

- **Same Direction:** Relative speed $= \mathbf{x - y}$.

$$(x - y) \cdot t_{\text{same}} = D$$

- **Opposite Directions (Towards each other):** Relative speed $= \mathbf{x + y}$.

$$(x + y) \cdot t_{\text{opp}} = D$$

Worked Archetype 10 | Taxi Cab Fixed and Distance Charges

Problem: The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is ¤105 and for a journey of 15 km, the charge paid is ¤155. What are the fixed charges and the charge per km? How much does a person have to pay for travelling a distance of 25 km?

Step 1: Formulate linear equations:
Let fixed charge $=$ ¤x, and charge per km $=$ ¤y.

$$x + 10y = 105 \quad \text{— (1)}$$

$$x + 15y = 155 \quad \text{— (2)}$$

Step 2: Subtract Equation (1) from Equation (2):

$$(x + 15y) - (x + 10y) = 155 - 105 \implies 5y = 50 \implies \mathbf{y = 10}$$

Step 3: Substitute $y = 10$ into Equation (1):

$$x + 10(10) = 105 \implies x + 100 = 105 \implies \mathbf{x = 5}$$

$$\text{Fixed Charge} = ¤\mathbf{5}, \qquad \text{Charge per km} = ¤\mathbf{10}$$

Step 4: Compute charge for 25 km:

$$\text{Total Cost} = x + 25y = 5 + 25(10) = 5 + 250 = ¤\mathbf{255}$$

CHAPTER 03

QUICK REFERENCE

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 11 | Master Formula Matrix & Solution Decision Protocol

11.1 The Comprehensive Linear Systems Matrix

Method / Criterion	Prerequisite Form	Mathematical Rule	Strategic Board Examination Purpose
Ratio Test	Standard form	$\frac{a_1}{a_2}$ vs $\frac{b_1}{b_2}$ vs $\frac{c_1}{c_2}$	Pre-test consistency before solving.
Graphical Method	Table of values	Intersection (x_0, y_0)	Visual proof; finding bounded triangle area.
Substitution	Isolate 1 variable	$x = f(y) \implies a_2 f(y) + b_2 y = c_2$	Best when a variable has coefficient ± 1.
Elimination	Equalize coefficients	Add if diff signs, Sub if same	Fastest general method for non-unit coefficients.
Symmetric Shortcut	$ax + by = c, bx + ay = d$	Add for $x+y$, Sub for $x-y$	Eliminates huge multi-digit LCM calculations.
Triangle with x-axis	Intersecting lines	Vertices $P(x_0, y_0), A, B$	Area $= \frac{1}{2} \cdot \lvert x_B - x_A \rvert \cdot \lvert y_0 \rvert$.
Number Reversal	2-digit number	$10x + y \leftrightarrow 10y + x$	Sum is $11(x + y)$; Difference is $9(x - y)$.

11.2 The 3-Tier Linear System Decision Flowchart



Strategic Rules for Method Selection:

- **Use Substitution when:** At least one variable in either equation has coefficient $+1$ or -1 (e.g., $x + 2y = 3$).
- **Use Elimination when:** All coefficients are > 1 and non-symmetric (e.g., $9x - 4y = 2000$ and $7x - 3y = 2000$).
- **Use Add/Subtract Shortcut when:** Coefficients are large and interchanged ($152x - 378y$ and $-378x + 152y$).

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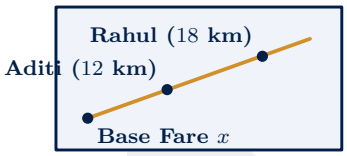
PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 12 | Applied Case-Study: Urban Transport & Electric Cab Pricing

Case Study 1 | Eco-Transit Smart Mobility Fare Structure

A metropolitan city introduces an electric green-cab service. The pricing policy consists of a minimum initial boarding fee (fixed charge) plus a constant variable rate per kilometer.

- Aditi travels 12 km from the railway station to her office and pays ¤145.
- Rahul travels 18 km across the city for a business summit and pays ¤205.



Cost Model: $C = x + dy$

4-Mark Board Question Protocol

Questions:

1. Represent the given information algebraically as a pair of linear equations in two variables. [1M]
2. Find the fixed charge and the charge per kilometer. [2M]
3. How much will a commuter pay for a journey of 30 km? [1M]

Solution & Marking Scheme Evaluation

1. **Algebraic Formulation [1M]:** Let fixed boarding charge = ¤x, and rate per km = ¤y.

$$x + 12y = 145 \quad \text{— (1)}$$

$$x + 18y = 205 \quad \text{— (2)}$$

2. **Solving for x and y [2M]:** Subtract Equation (1) from Equation (2):

$$(x + 18y) - (x + 12y) = 205 - 145$$

$$6y = 60 \implies \mathbf{y = 10}\ ¤/\text{km}$$

Substitute $y = 10$ into Equation (1):

$$x + 12(10) = 145 \implies x + 120 = 145 \implies \mathbf{x = 25}\ ¤$$

Answer: Fixed charge is ¤**25**, and charge per km is ¤**10**.

3. **Fare for 30 km Journey [1M]:**

$$\text{Total Fare} = x + 30y = 25 + 30(10) = 25 + 300 = ¤\mathbf{325}$$

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CHAPTER 03

EXAM STRATEGY

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

13.1 The Top 7 Linear Equation Board Traps (And How to Bust Them)

Trap 1: The Constant Term Position Asymmetry Trap: Comparing $\frac{c_1}{c_2}$ when one equation has constant on LHS and other on RHS (e.g., $2x + 3y = 7$ and $4x + 6y - 14 = 0$).

TRAP ALERT

You must move both constants to the **same side** before taking ratios!

Trap 2: The Missing Sign in Subtraction of Equations: In elimination, when subtracting $(-3y)$ from $(-5y)$, mistakenly writing $-8y$ instead of $-5y - (-3y) = \mathbf{-2y}$.

Trap 3: The Incomplete Parameter k Check: In infinite solutions, taking only $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ and forgetting to check $\frac{b_1}{b_2} = \frac{c_1}{c_2}$. A single equation cannot confirm infinite solutions!

Trap 4: The Two-Digit Number Variable Trap: Writing the two-digit number as xy or $x + y$. The decimal number is strictly $\mathbf{10x + y}$.

Trap 5: Shading the Wrong Triangular Region: Shading the triangle formed with the y-axis when the question asks for the triangle formed with the **x-axis**. Always highlight the specified axis first!

Trap 6: Inverting Relative Speed in Rivers: Writing upstream speed as $y - x$. The boat speed must exceed stream speed, so upstream is strictly $\mathbf{x - y}$.

Trap 7: Using the Deleted Cross-Multiplication Method: The cross-multiplication formula is deleted from the CBSE curriculum. Solve strictly using **Substitution** or **Elimination**.

13.2 High-Speed Algebraic Verification Protocol

Before turning in your answer sheet, always substitute your computed values of (x, y) into **both original equations**:

$$\text{Check 1: } a_1x_0 + b_1y_0 + c_1 \stackrel{?}{=} 0 \qquad \text{and} \qquad \text{Check 2: } a_2x_0 + b_2y_0 + c_2 \stackrel{?}{=} 0$$

If it satisfies both, your solution is 100% mathematically guaranteed!

13.3 The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Verification Protocol:

- ☐ **Standard Form Alignment:** Were both equations rearranged with constants on the same side before checking ratios?
- ☐ **Ratio Comparison Stated:** Did you write $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ explicitly before declaring unique solution? (Guarantees method marks).
- ☐ **Both Variables Solved:** Did you find both x AND y? (Don't leave after finding just one!).
- ☐ **Triangle Vertices Explicitly Listed:** In graphical questions, did you write the coordinates of all 3 vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$?
- ☐ **Units Attached in Word Problems:** Incomes in ¤, Speeds in km/h, Ages in years!

MATHS MANTHAN TUITION CLASSES | END OF CHAPTER 3 HANDBOOK

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